

34721/34722 Linear Control Design 1

Spring Semester



Silvia Tolu & Dimitrios Papageorgiou

Lesson 1/13

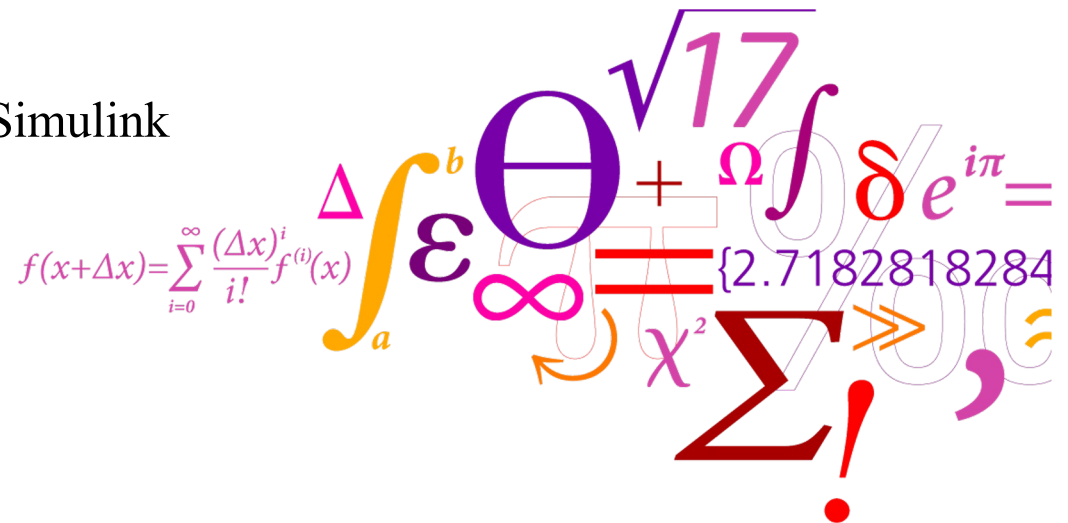
Schedule

13:00 -15:00

- Welcome and intro to the course
- Pre-test

15:00 - 17:00

- Exercises in Matlab and Simulink



About the lecturer



2023-	Head of Neurorobotics Technology Lab (NRT-LAB)
2021 -	Associate Professor in Neurorobotics
2019-2020	Assistant Professor at DTU Electro
2015-2019	Postdoc at DTU Electro
2014	Robotics Consultant in Genoa (Italy)
2012	PhD in Neuromorphic computing from the University of Granada (Spain)
2006	M.Sc. in Electronic Engineering from the University of Cagliari (Italy)



Silvia Tolu



stolu@dtu.dk



Building 326/106

About the lecturer



2023 – present	Associate Professor in Nonlinear and Fault-tolerant Control	
2020 – 2023	Assistant Professor of Automation and Control	
2017 – 2020	Postdoc	
2017	PhD in “High-accuracy nonlinear control and estimation for machine tool systems”	
2014	M.Sc. in Electrical Engineering	
2011	B.Sc. Degree and combined M.Sc. Equivalent in Electrical and Computer Engineering	



Dimitrios Papageorgiou



dimpa@dtu.dk



Building 326/124

Teachers - Course Responsible

Associate Prof. Dimitrios Papageorgiou (B. 326, R. 124), Associate Prof. Silvia Tolu (B. 326, R. 106)



Teaching Assistants (TAs)



Hans Christian



Dimosthenis



Kornel



Susan



Eduard

TAs will provide feedback on the assignments during exercise time
(for 34721 in B329A-IT120 (Agora building) and for 34722 in B325-H009/H017).

Use *Learn* Discussion forum to have *questions & answers* for everyone.

TEXTBOOK

Feedback control Techniques by Andersen, Santos, Niemann. Available in DTU bookshop

https://polyteknisk.dk/home/dtu/soege_resultat?utf8=%E2%9C%93&q=34722&b=20&st=c&code=&exact_match=false

Hardware issues? → Contact: Søren Beyer Nielsen, sbeni@dtu.dk

Jens Christian Andersen, jcan@dtu.dk

Lectures are held in **auditorium A021 in B341** (DTU Lyngby Campus).

Group work is mostly project work and exercises (all in groups).

Additionally, there will be discussion topics in DTU Learn (under discussions) where you can ask questions and clarifications about the exercises.

What about the exam?

The exam is a 4-hour written exam, supplemented by a report on the last assignment (overall assessment, 7-point scale).

Will the course be hard?

Yes, probably a little.

Will it be exciting and interesting?

Yes, at least we hope so.

Course day

(4 hours / Wednesday 13:00 - 17:00)

Overall schedule

Lecture with breaks (13:00 – 15:00)

Groupwork (15:00 - 17:00)

- Discussion on today's exercise
- Work on assignments

After the course day

- Online quizzes will be available in DTU Learn for practice

What is Linear Control Design 1 about?



What is a control system?

A **control system** is a system whose behavior needs to be regulated. Typical examples include:

- the speed of an engine
- the water level or temperature in a tank
- the pH value in a bioreactor
- the frequency generated by a power plant
- the sound pressure produced by a loudspeaker

For a system to be controlled, it must have at least **one control input**.

This input may be, for example, the voltage applied to a motor, the opening position of a valve, or the power supplied to a heater.

What is a control action?

The term **control action** refers to the way a **control input is generated**.

In feedback control, the output of the system is measured and compared with a desired value (the **reference**). The difference between the measured value and the desired value, called the **error**, is used to determine the control signal in a closed-loop system.

For example, the valve position may be adjusted based on the measured water level in a tank.

What is a control object?

A **control objective** defines the tasks to be performed and the goals to be achieved to obtain the desired system performance. This may include requirements such as **stability, accuracy, speed of response, or robustness to disturbances**.

What is Linear Control Design 1 about?

Linear Control Design 1 is about the design of PID-based controllers.

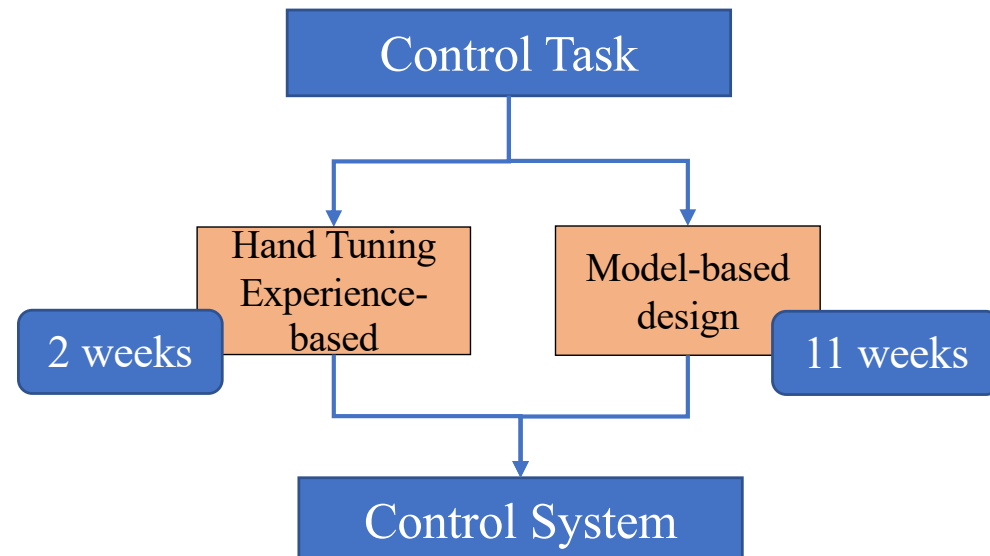
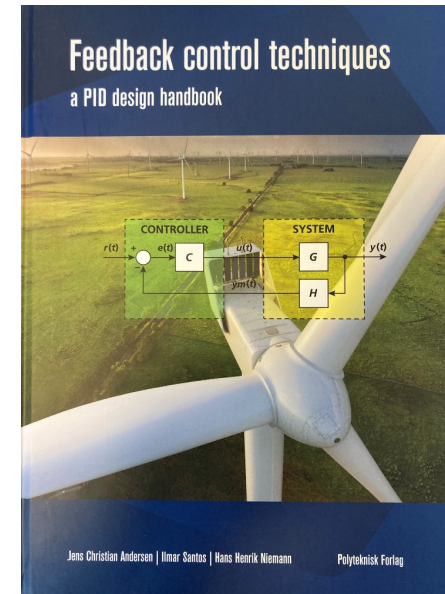
PID stands for **Proportional**, **Integral** and **Differential**.

The main goal is to find the optimal values for

$$K_P, \tau_i, \tau_d$$

to determine the control action $u(t)$ from the error $e(t)$

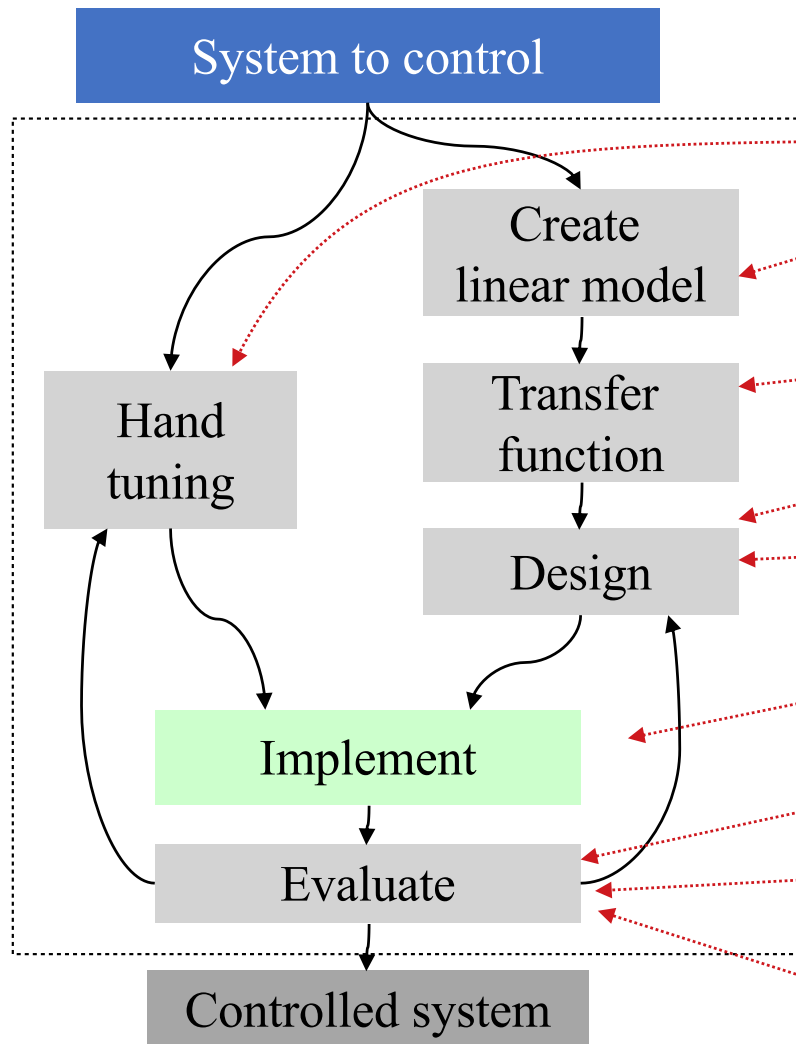
$$u(t) = K_P \left(e(t) + \frac{1}{\tau_i} \int_0^t e(t) + \tau_d \frac{d}{dt} e(t) \right)$$



Linear control design (34721/34722)



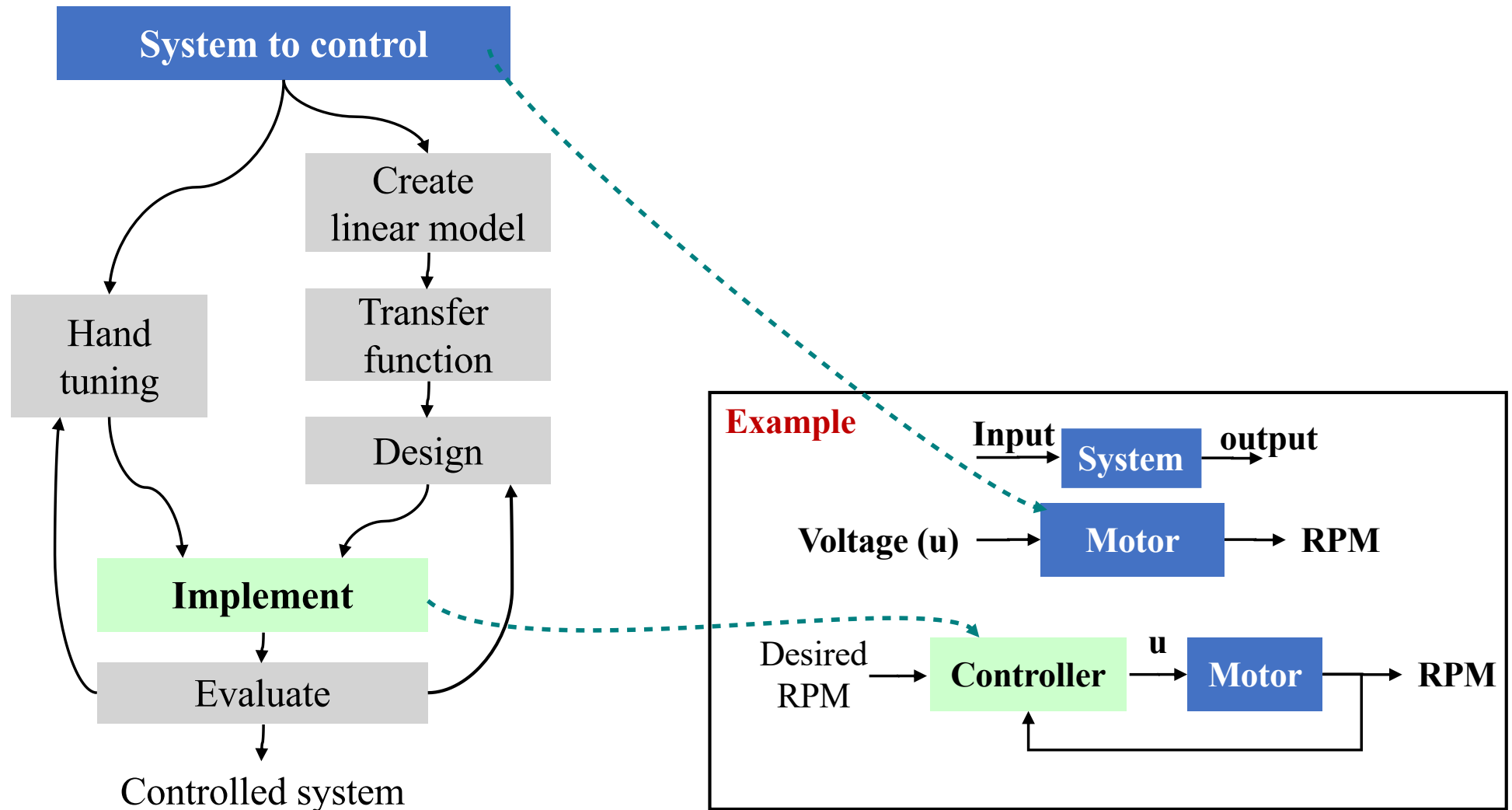
Learning Objectives



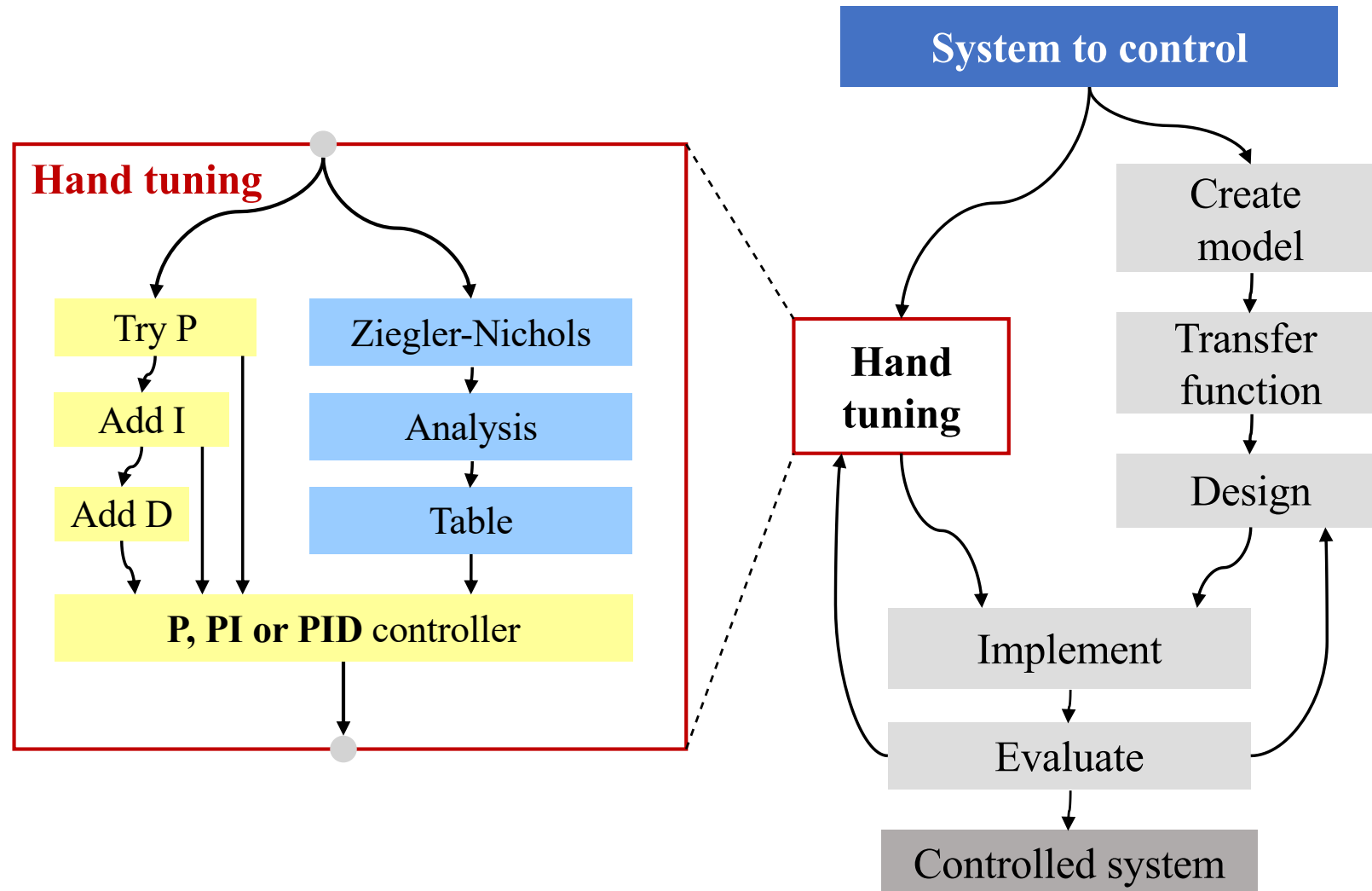
A student who has met the objectives of the course will be able to:

- 1) **Apply** heuristic methods for PID controllers.
- 2) **Derive** models of simple dynamic systems based on physics or measurements and linearize the system to a transfer function in a single operating point.
- 3) **Analyze** transfer functions in the frequency domain (Bode plot).
- 4) **Apply** block diagrams to describe models for both open- and closed-loop control systems.
- 5) **Design** PID controllers based on open loop frequency analysis.
- 6) **Apply** the concepts of stability and stability margin from open loop design.
- 7) **Understand** the consequences of disturbance and noise sensitivity.
- 8) **Evaluate** controller performance in terms of overshoot, settling time, steady state error, bandwidth and control limitations.
- 9) **Apply** analysis, design and simulation software.

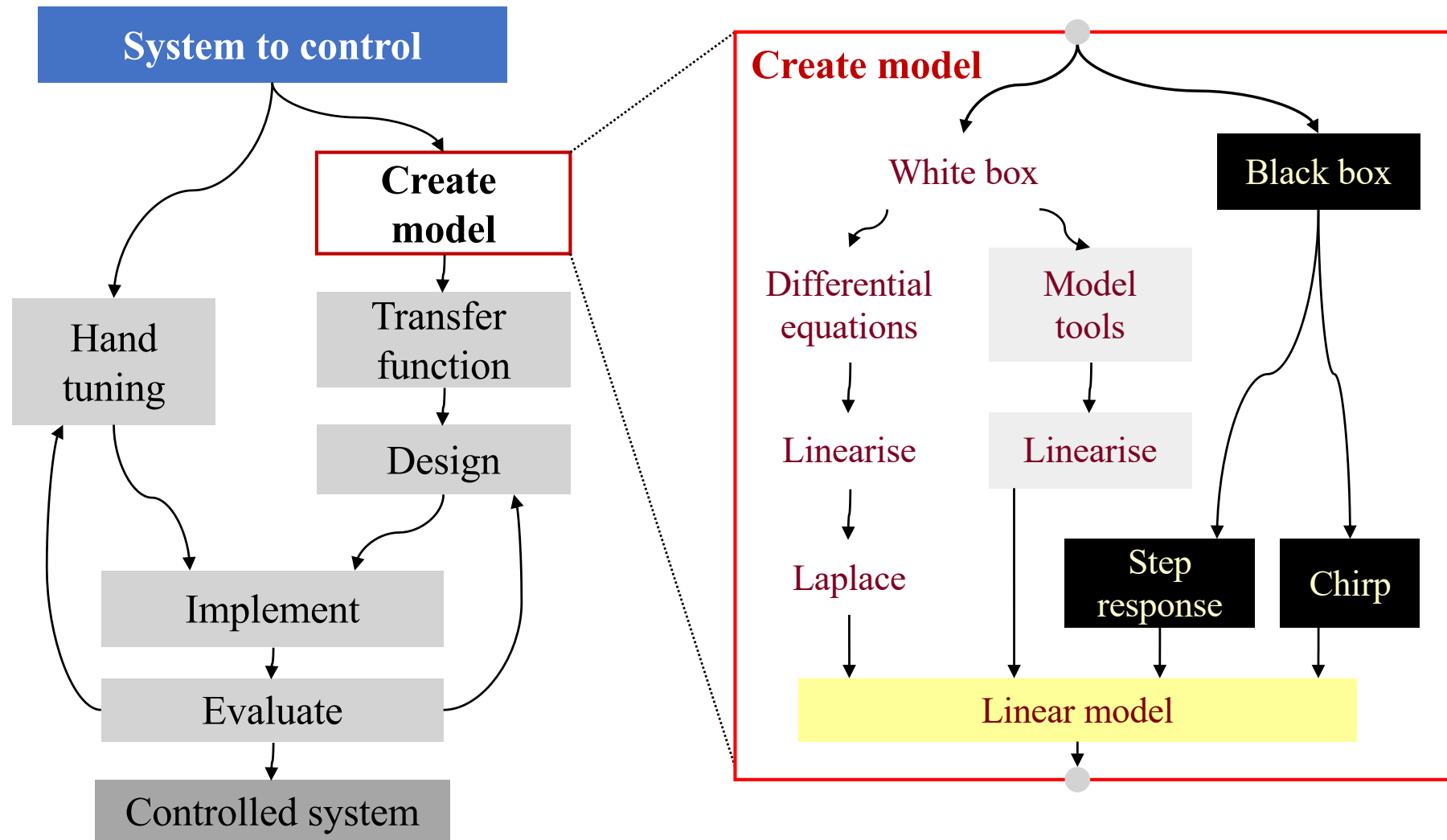
Linear control design



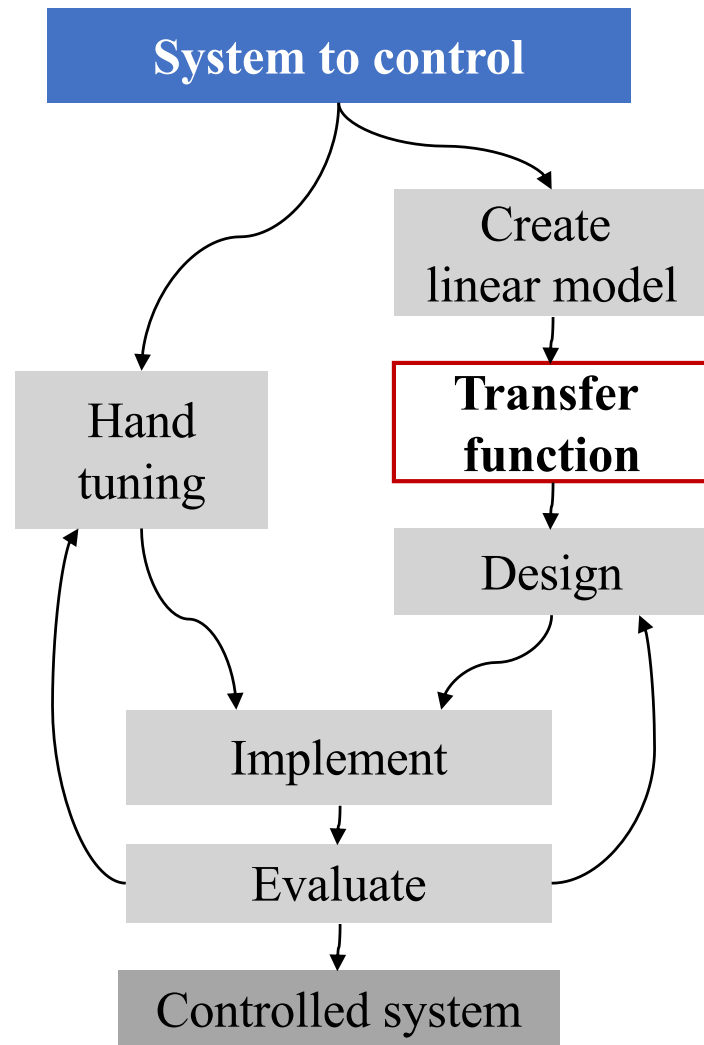
Linear control design



Linear control design



Linear control design

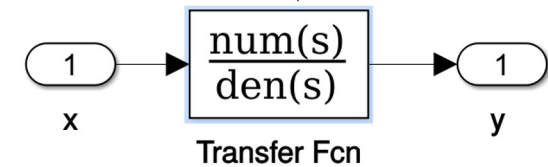


Linear model

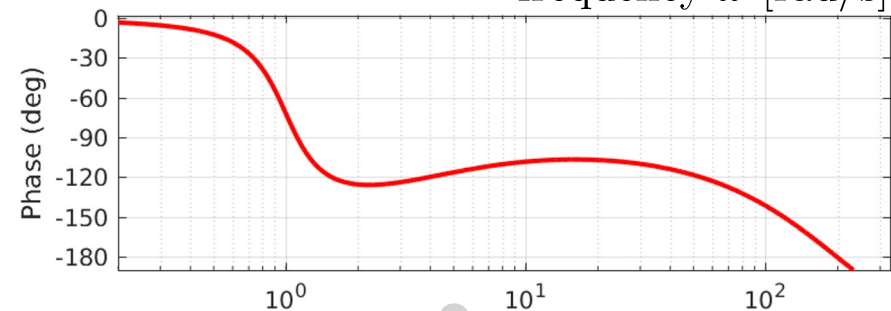
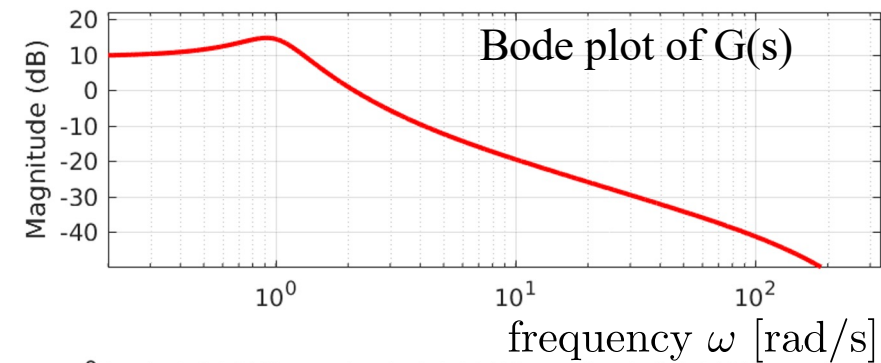
Laplace
Transfer Function

$$y(s) = G(s)x(s)$$

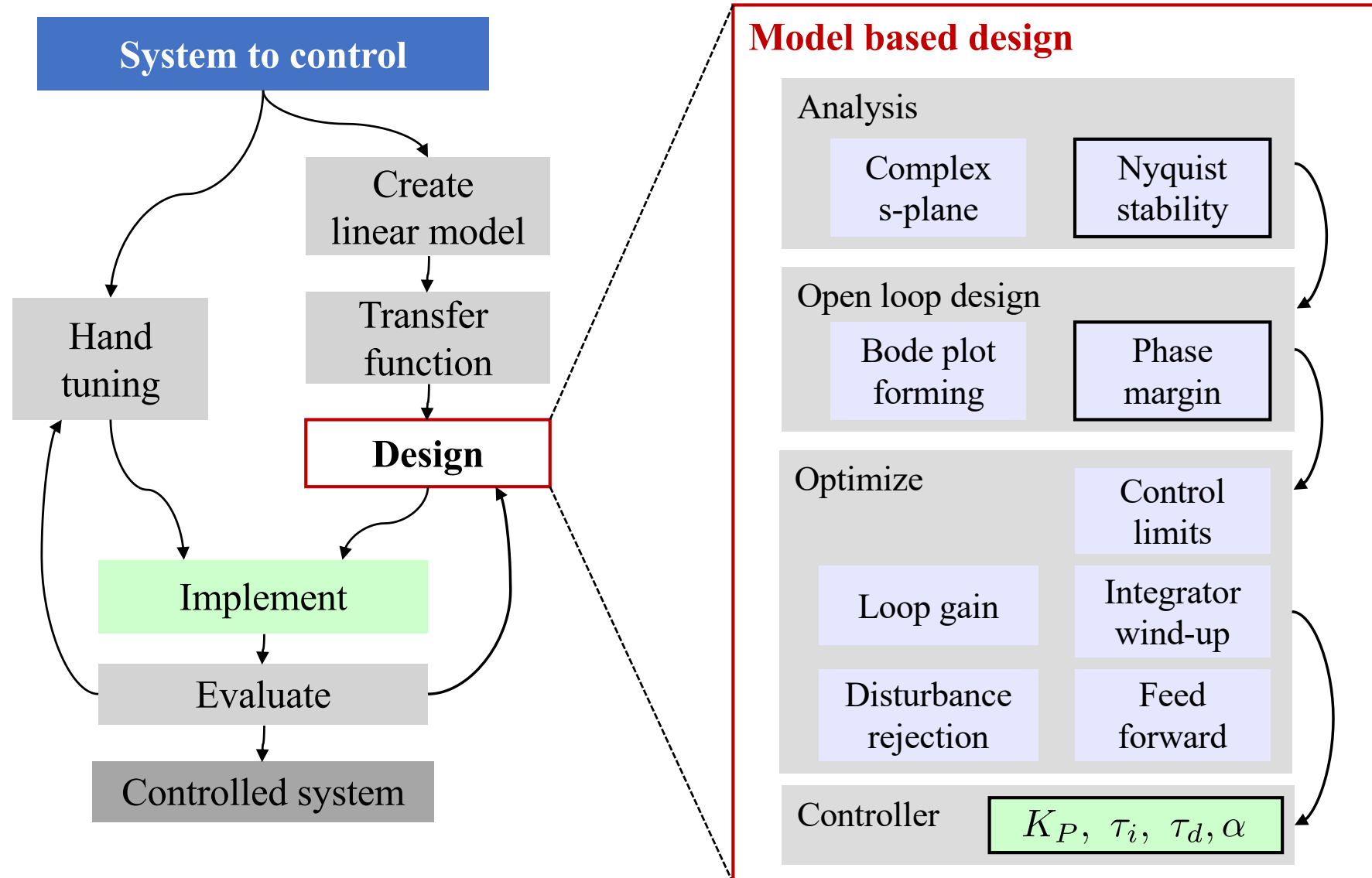
$$s = \sigma + j\omega$$



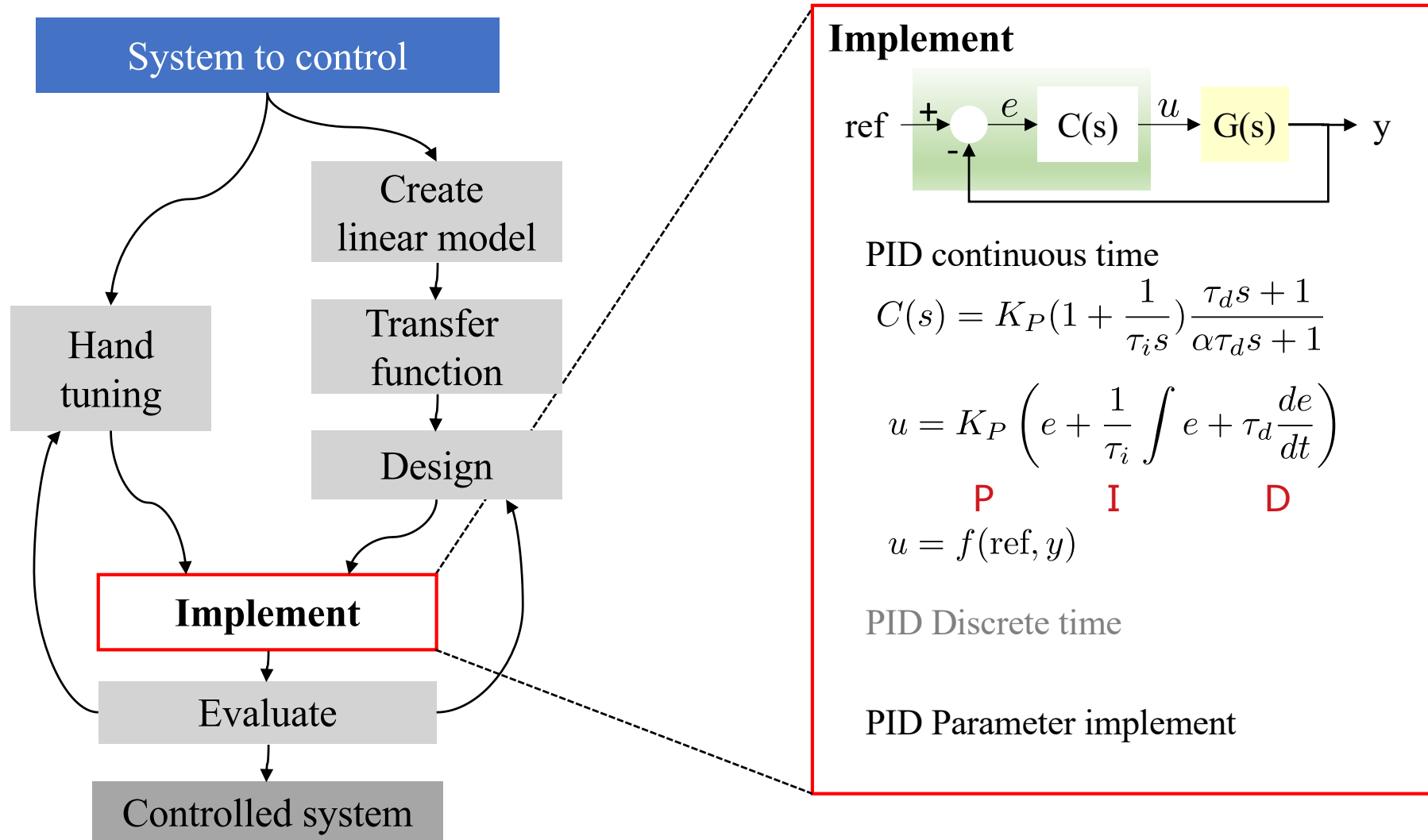
$$G(s) = \frac{s + 3}{250000s^4 + 0.01s^3 + s^2 + 0.61s + 1}$$



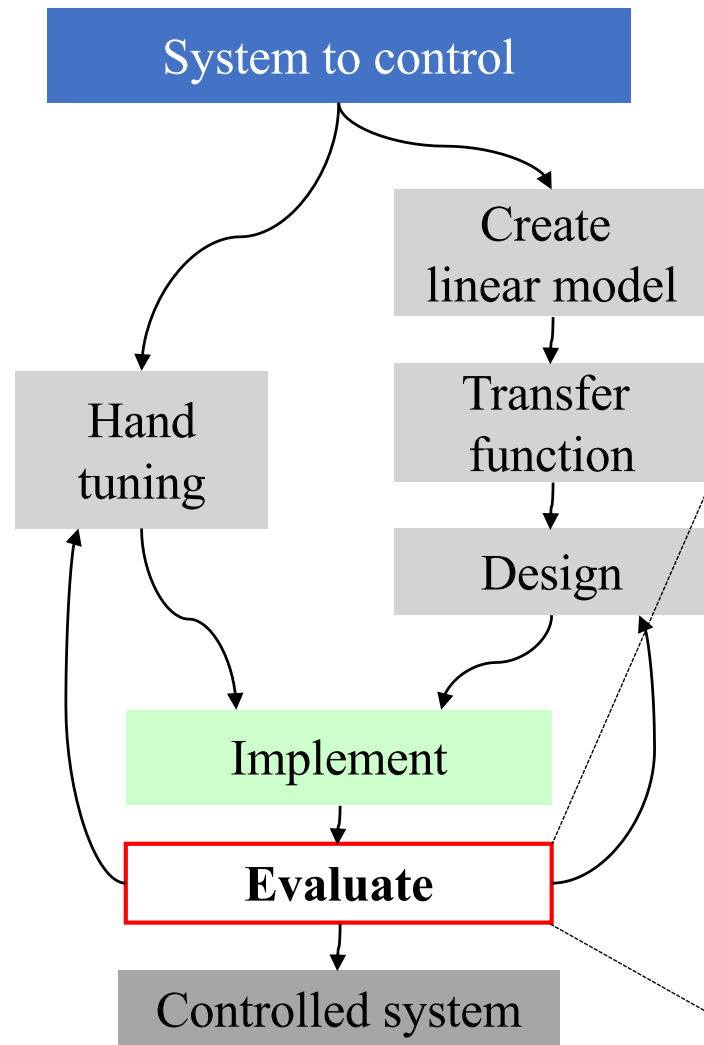
Linear control design



Linear control design

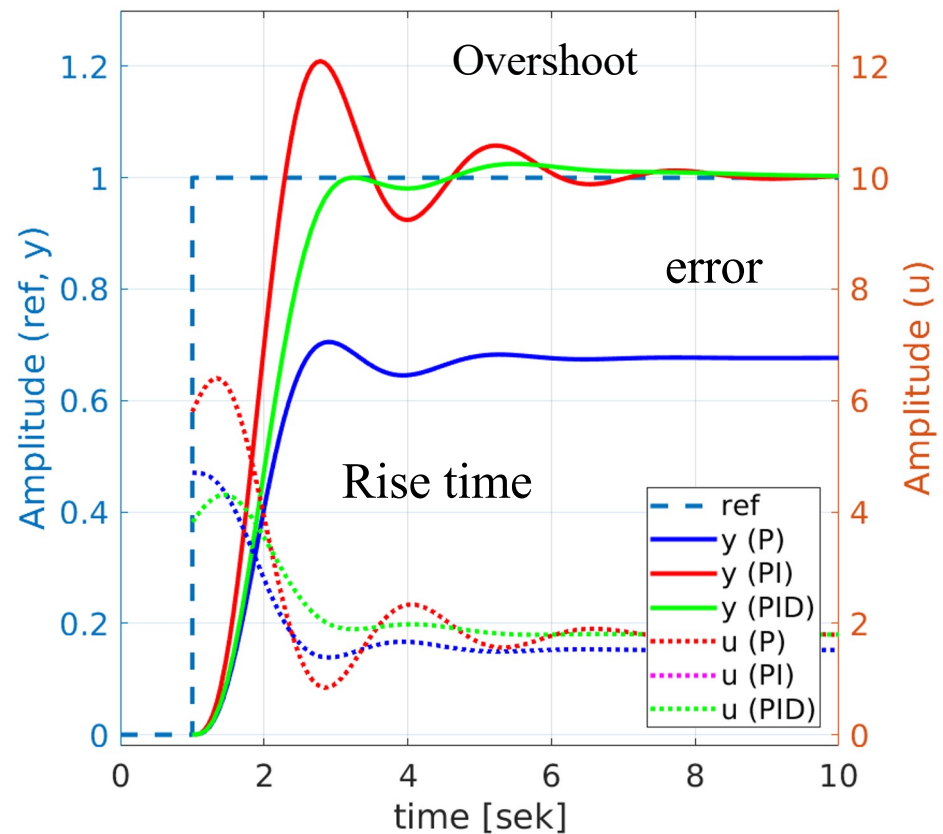
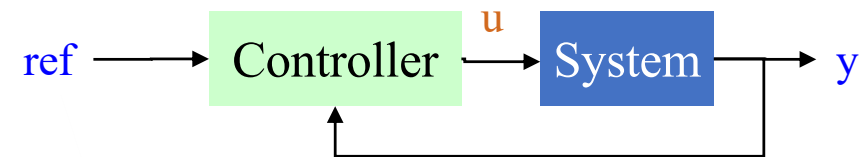


Linear control design



Evaluate

Target: $y = \text{ref}$



Course Content



#	Topics
1	Introduction
2	Control concepts
3	Transfer function and Laplace
4	Frequency domain
5	Modelling and linearization
6	Bode plots and stability
7	Mid-term test , stability criteria
8	PI-Lead Design - Part 1
9	PI-Lead Design - Part 2
10	Unstable systems
11	Limited systems
12	Disturbances, sensitivity and pre-filters
13	Feed-forward control

#	Topics	Assignments
1	Introduction	MATLAB and Simulink
2	Control concepts	Hand tuning
3	Transfer function and Laplace	Block diagrams and transfer functions
4	Frequency domain	Introduction to REGBOT
5	Modelling and linearization	REGBOT – Transfer function from data
6	Bode plots and stability	REGBOT – Bode, stability, P-Controller design
7	Mid-term test , stability criteria	No new assignment (lecture instead)
8	PI-Lead Design - Part 1	Controller design (speed, heading, position)
9	PI-Lead Design - Part 2	Controller design (speed, heading, position)
10	Unstable systems	REGBOT balance
11	Limited systems	REGBOT balance
12	Disturbances, sensitivity and pre-filters	REGBOT balance
13	Feed-forward control	REGBOT balance – Final report

Final Report (mandatory) includes work done in weeks 10-13

Requirements for the Exam



To take the 4-hour written exam:

- **You have to submit the final report based on the last mandatory (REGBOT) assignment for the current semester**
 - It is 5 pages max. The instructors and an external censor will evaluate it. It will count in the final grade and cannot be skipped.
- The grade is an overall assessment (helhedsvurdering) of the final report **and** the final written exam.

Re-use of the report: The report on the mandatory assignment can only be used for **two exams attempts**, i.e., the regular exam in the same semester as the assignment and the first upcoming re-exam. (*Example 1:* A student takes the regular exam in Fall and the re-exam in Spring; *Example 2:* A student takes the regular exam in Spring and the re-exam in August). A report can be used for the re-exam **ONLY** if the assessment is PASS.

When the report is used for the re-exam, its assessment **DOES NOT** contribute to the final grade (only the re-exam score itself counts). **STRICT** deadlines (announced in class and on DTU Learn course page) apply to inform the lecturers if the student wishes to use an existing report for the re-exam. The re-exam may be written or oral; the format will be announced by the lecturers.

Forming groups and getting the REGBOT



The assignments are made in **Groups of 4 students**

- Find your group mates and join the group on LEARN (“My course”, “groups”, “view available groups”, “join”)

The REGBOT can be picked up in building 326

- The exact dates and office will be announced
- Fill out the “**group form**” (excel file in TAs office) with the robot number and your group number

DAY 1 Exercise

- 
- **Watch videos** about introduction to **MATLAB and Simulink**
 - Do the exercises about MATLAB and Simulink

Pre-Test (1h)